How many different materials are there?

1 Materials used in a modern car include steel, glass, plastic, rubber, lead and copper.

Choose a suitable material for each of the following parts of a car and suggest why it is suitable for this use.

a electrical wiring

b windscreen

c car body

d tyres

e steering wheel

f plates in a car battery

2 Today, many things we use every day are made of some type of plastic. Most of these plastics are made from crude oil. When your grandparents were young they did not have all these different plastics. They had to use other materials.

a Suggest materials that were used for the following before plastics were discovered:

i washing up bowl

ii wrapping up sandwiches

iii insulation for electrical wires

iv milk crates

v car bumpers.

b In 2000 the Nobel Prize for Chemistry was awarded to scientists who invented a plastic that conducts electricity. Suggest why this could be useful in the future.

3 About 500 years ago people called alchemists tried to change metals such as lead into gold. This would involve changing one element into another.

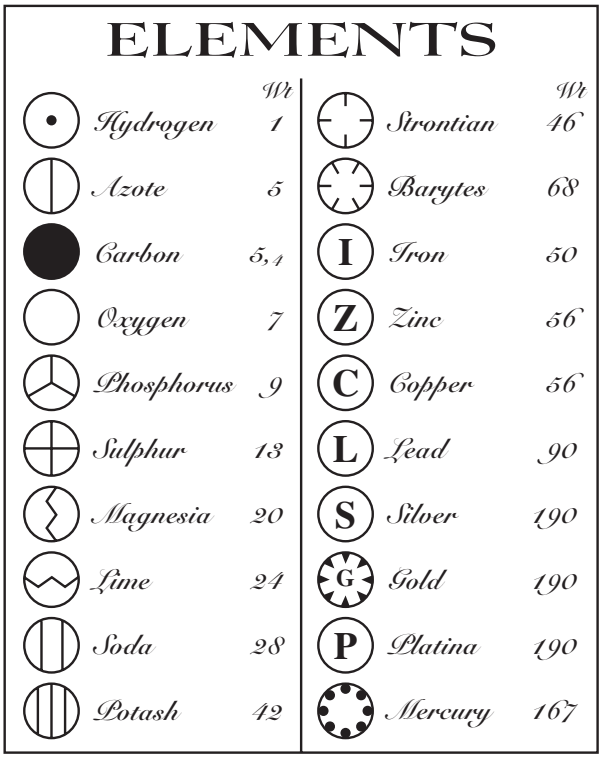
a Why would they want to do this?

b If it was possible to do this, a lot of gold would be made. What would happen to the price of gold?

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What are elements made from?

In 1808, John Dalton produced a list of the elements he knew. He gave these elements symbols. Here is his list.



1 How many elements did Dalton list?

2 Which element did he represent as  ?

3 He used his symbols to represent substances made from elements.
What do you think he meant by  ?

4 Some of the substances he thought were elements we now know are not elements.

a Which substances in his list are elements? The Periodic Table should help you.

b Which substances in his list do you think are not elements?

5 Write down three metallic elements in Dalton's list and three non-metallic elements.

6 Today we use symbols made up of one or two letters to represent the elements. Why are these symbols easier to use than Dalton's symbols?

			H hydrogen														He helium
Li lithium	Be beryllium												B boron	C carbon	N nitrogen	O oxygen	F fluorine	Ne neon
Na sodium	Mg magnesium												Al aluminium	Si silicon	P phosphorus	S sulphur	Cl chlorine	Ar argon
K potassium	Ca calcium	Sc scandium	Ti titanium	V vanadium	Cr chromium	Mn manganese	Fe iron	Co cobalt	Ni nickel	Cu copper	Zn zinc	Ga gallium	Ge germanium	As arsenic	Se selenium	Br bromine	Kr krypton	
Rb rubidium	Sr strontium	Y yttrium	Zr zirconium	Nb niobium	Mo molybdenum	Tc technetium	Ru ruthenium	Rh rhodium	Pd palladium	Ag silver	Cd cadmium	In indium	Sn tin	Sb antimony	Te tellurium	I iodine	Xe xenon	
Cs caesium	Ba barium	La lanthanum	Hf hafnium	Ta tantalum	W tungsten	Re rhenium	Os osmium	Ir iridium	Pt platinum	Au gold	Hg mercury	Tl thallium	Pb lead	Bi bismuth	Po polonium	At astatine	Rn radon	
Fr francium	Ra radium	Ac actinium																

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What are elements like?

Here is some data for ten elements.
The elements are given letters A–J.
These are not the chemical symbols.
Use this information to help you answer the questions.

Element	Melting point (°C)	Boiling point (°C)
A	−219	−183
B	1539	2887
C	−39	357
D	−7	58
E	419	908
F	119	445
G	1083	2582
H	−241	−196
I	114	183
J	78	890

1 Which element in the table has the lowest melting point?

2 Which element in the table has the highest boiling point?

3 Which elements are:
a solids
b liquids
c gases
at room temperature (20 °C)?

4 Which element is liquid over the greatest range of temperature?

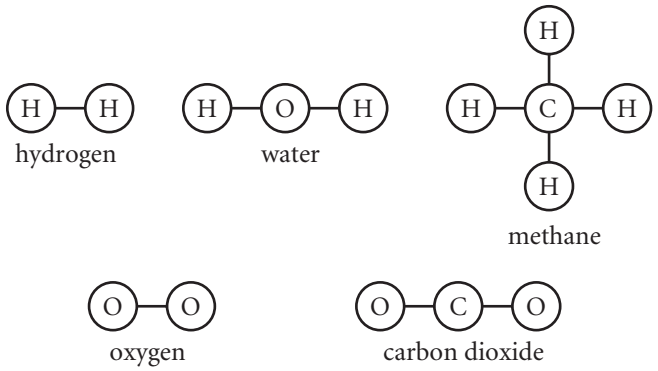
5 In Siberia, the winter temperature is −40 °C. Which elements would be in a different state at this temperature than they would be at normal room temperature?

6 On a distant planet the temperature is −160 °C. Why could there be no element A in the atmosphere of this planet?

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How do we get all the other materials?

1 The diagram shows some molecules.

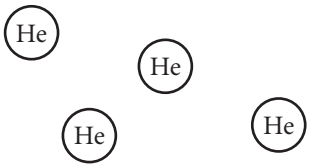


Make a table with these headings:

Substance	What types of atom are in the molecule?	Is the substance an element or a compound?

Complete your table for the substances in the diagram.

2 The diagram opposite shows helium.

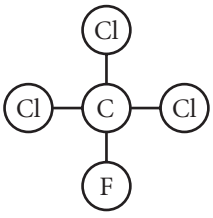


a Does helium have molecules? Explain your answer.

b Is helium an element or a compound?

3 Chlorofluorocarbons are compounds that used to be used in fridges. They cause problems when they escape as gases into the atmosphere.

A molecule of chlorofluorocarbon can be shown as:

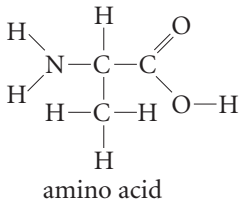


chlorofluorocarbon

a Write down the names of the three elements in this compound.

b How many atoms of each element are there in a molecule of this compound?

4 The diagram opposite shows a simple amino acid, a product of food digestion.

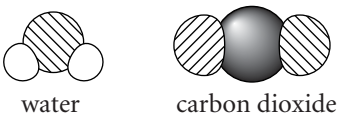


amino acid

a What four elements are combined in this compound?

b How many atoms altogether are there in a molecule of this compound?

5 Molecules are often shown in two-dimensional drawings. These show the way the different atoms are joined, but not necessarily how they are arranged in space. These diagrams show the shapes of the molecules.


water carbon dioxide

Describe the shape of a molecule of:

a water **b** carbon dioxide.

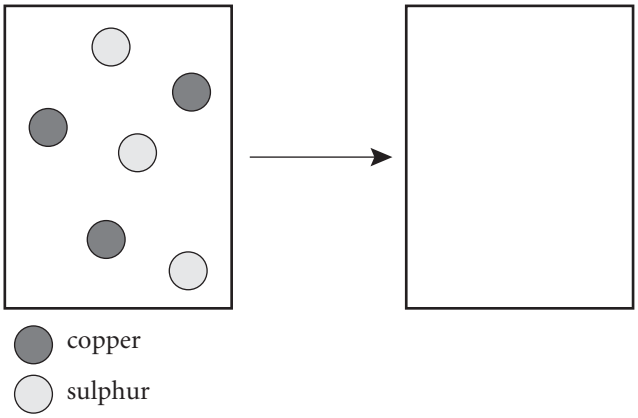
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How can we record the changes when atoms join?

1 Copper and sulphur combine to form copper sulphide.

a Write the word equation for this reaction.

b Copy and complete this diagram showing the atoms combining.



2 When hydrogen and oxygen combine, molecules of water are produced. These molecules contain two hydrogen atoms and one oxygen atom.

In both gases hydrogen and oxygen, the hydrogen and oxygen are present as molecules. Each molecule contains a pair of atoms.

a Write a word equation for the reaction.

b Draw diagrams under your word equation showing the three sorts of molecule. Use red for oxygen and white for hydrogen.

c The properties of hydrogen and oxygen gases are different from the properties of the compound water which is formed when they combine.

Describe some properties of hydrogen, oxygen and water.

3 When compounds are formed, energy is often released to the surroundings. From your experience, in which form is this energy released?

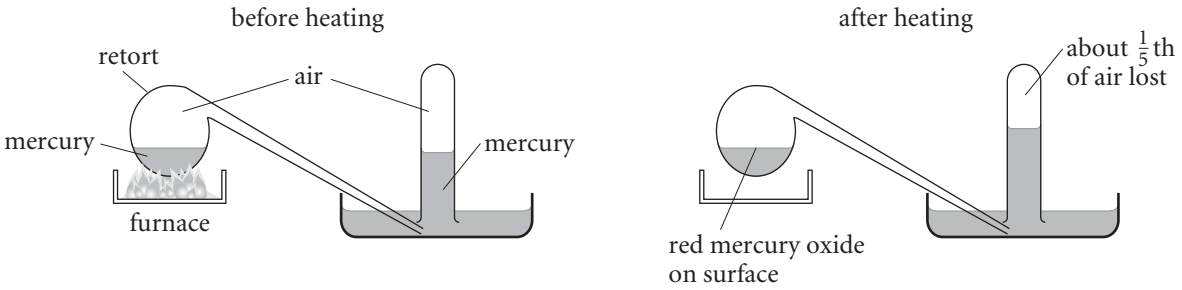
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Getting ahead of the game

Lavoisier

Antoine Laurent Lavoisier was a famous scientist in France working at the end of the eighteenth century. He made an important contribution to explaining what happens when things burn.

He heated mercury in a certain volume of air. After the experiment he found some red powder on the surface of the mercury in the retort. The diagrams show his apparatus before and after heating.



Lavoisier carefully removed the red powder. He heated it and collected a colourless gas in which things burned better than in air.

1 **a** Write down the name of the gas Lavoisier collected.

b Use your understanding to explain what was happening during Lavoisier's experiment.

c Write a word equation for the reaction taking place when he heated mercury in air.

d Write a word equation for the reaction taking place when he heated the red powder.

2 Lavoisier found that:

- mercury **gained** weight when heated in air to form an oxide
- the oxide **lost** weight when it was heated.

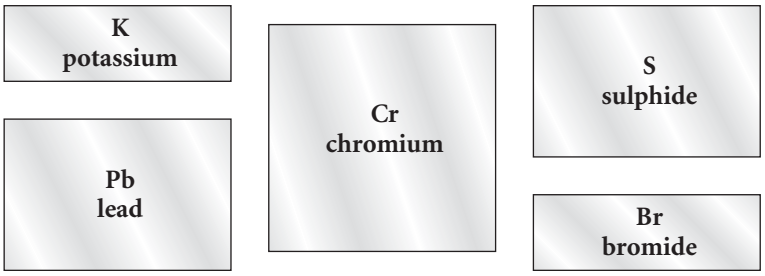
Until then scientists believed that when a substance burned it **lost** a substance called phlogiston.

Explain why Lavoisier's work meant the end of the previous theory of combustion.

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All about elements and compounds

1 Write the formulae for the compounds below. Use the idea of cards making rectangles to help you, as in the diagram.



a potassium bromide **b** potassium sulphide **c** lead sulphide
d lead bromide **e** chromium bromide **f** chromium sulphide

2 Copy and complete the table.

Compound	Formula	Number of atoms present
calcium chloride	$CaCl_2$	one atom of calcium, two atoms of chlorine
sulphuric acid	H_2SO_4	
calcium hydroxide	$Ca(OH)_2$	
sodium carbonate	Na_2CO_3	
sodium hydrogencarbonate	$NaHCO_3$	
		One atom of magnesium, one atom of sulphur, four atoms of oxygen

3 Sulphur dioxide is made by burning sulphur in oxygen.

Five students made sulphur dioxide, each using different masses of sulphur. The table shows the mass of sulphur used and the mass of sulphur dioxide produced.

Student	Mass of sulphur (g)	Mass of sulphur dioxide (g)
1	0.60	1.20
2	0.70	1.40
3	0.80	1.30
4	0.90	1.80
5	1.00	2.00

Which student's results are most likely to be wrong? Explain your reasoning.

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How do compounds differ from the elements in them?

1 Carbon and oxygen can form two oxides:

- carbon dioxide, CO_2
- carbon monoxide, CO

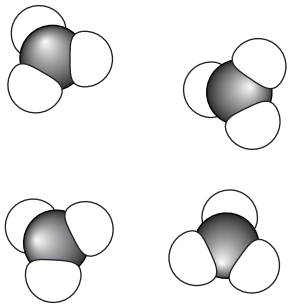
They are both gases.

a Which gas contains molecules made up of only two atoms?

b What is the ratio of carbon atoms to oxygen atoms in each oxide?

c How is a molecule of carbon monoxide different from a molecule of oxygen, O_2?

2 **a** Does the diagram represent an element or a compound? Explain your choice.



b What is the ratio of the atoms of different elements in the molecules shown?

3 **Hydrogen** is a colourless, odourless gas that explodes when lit.

Chlorine is a greenish-yellow gas which bleaches litmus paper.

Hydrogen chloride is a colourless gas which turns damp blue litmus paper red.

a Which substances in **bold** above are made up of molecules containing only one type of atom? Which are made up of molecules containing two types of atom?

A mixture of elements has the same properties as the elements in the mixture.

b Decide whether each of the following statements about a mixture of chlorine and hydrogen is true or false.

- It turns blue litmus red.
- It bleaches red litmus
- It has no smell.
- It has a pale greenish-yellow colour.
- It is colourless.

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Do compounds react chemically?

Substances	Appearance before mixing	Change on mixing	Appearance at the end
calcium carbonate and hydrochloric acid	white solid and colourless liquid	fizzes, colourless gas given off	colourless solution
hydrochloric acid and sodium hydroxide solution	colourless solutions	colourless solution, solution is hot	colourless solution
copper oxide and water	black solid and colourless liquid	black solid and colourless liquid	black solid and colourless liquid
copper oxide and warm sulphuric acid	black solid and colourless liquid	blue solution	blue solution
sucrose and concentrated sulphuric acid	white solid and oily colourless liquid	solid turns yellow, brown and then black, gives off strong smelling gas and steam, gets very hot	black solid

1 The table gives some information about changes that take place when substances are mixed together.

a Which pair of substances do not seem to react?

b For each other pair of substances, suggest why the evidence in the table suggests a reaction is taking place.

2 There are many things that can be observed that suggest a reaction is taking place. Make a list of as many as you can.

3 Sodium sulphate is added to water in a beaker. The sodium sulphate disappears and a colourless solution remains. Tim says the sodium sulphate has dissolved and Tina says the sodium sulphate has reacted. Suggest how you could show them who was correct.

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Compounds and mixtures

1 Rock salt is a mixture of sodium chloride and also some other substances that are insoluble in water.

a Describe how a sample of pure sodium chloride could be made from rock salt.

b Suggest why rock salt is a much cheaper form of salt to put on an icy path than table salt.

2 Calcium carbonate is sold to schools in two forms: **laboratory grade** for normal use, and a special pure form called **analytical grade**.

Here is some information about the two forms.

Calcium carbonate		
Laboratory grade	1 kg	£3.00
Assay: min 98.5%		
Maximum limits of impurity		
Arsenic		0.0004%
Chloride		0.05%
Iron		0.02%
Lead		0.02%

Calcium carbonate		
Analytical grade	250 g	£7.50
Assay: min 99.5%		
Maximum limits of impurity		
Chloride		<0.001%
Iron		<0.001%

a 'Assay' means purity. What is the maximum impurity in each grade?

b Why is analytical grade calcium carbonate more expensive than laboratory grade?

3 The composition of air in a city is different from that in the countryside. This is possible because air is a mixture. Explain why it would be impossible if air was a compound.

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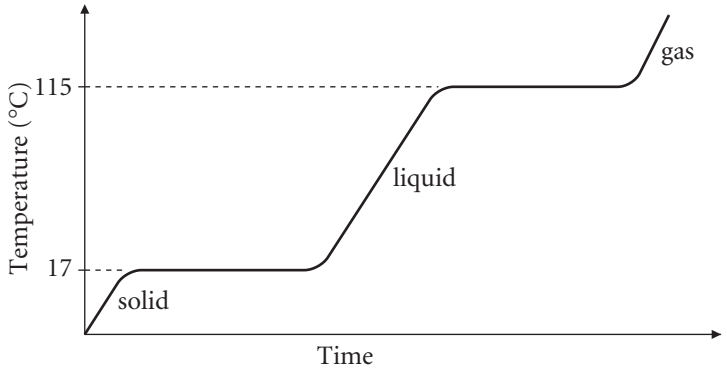
More differences between compounds and mixtures

1 The diagram below shows a graph obtained when a solid was heated at a steady rate while it changed first to a liquid and then to a gas. A graph like this is called a heating curve.

a What are the melting and boiling points of this substance?

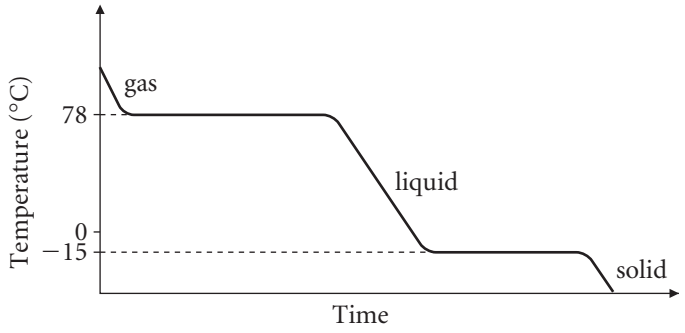
b What happens to the temperature when a change of state takes place?

c How can you tell this substance is not pure water?



2 Sketch the heating curve for pure water. Start with ice at −10 °C and finish with steam at 110 °C. Mark each change of state on the graph.

3 The cooling curve for a liquid is shown in the diagram below.



a What are the melting point and the boiling point of the substance?

b Is the substance a solid, liquid or gas at room temperature (20 °C)?

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Homework

G1

What are rocks made of?

1 Which of these is a mineral?

a granite

b sandstone

c calcium carbonate

d gabbro

2 Explain why a piece of sandstone is porous but a piece of granite is not.

3 Look in your garden or around where your home is. Find some rocks. Look carefully at each one. Use a magnifying lens if you have one.

Describe what each rock looks like.

Which rocks have an interlocking texture and which have a non-interlocking texture?.

Homework

G2

Rain and weathering

1 Which of the following does **not** cause physical weathering of rocks?

a waves

b wind

c frost

d acid

2 How are caves formed?

3 Explain why normal rainwater is slightly acidic.

4 Why do some weathered rocks have rust streaks?

5 Some rocks, such as limestone, are eventually weathered away completely by the action of rain. Other rocks, such as granite, are broken down into other materials. For example, granite forms clay and quartz. Explain this difference in the action of rain on different rocks.

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